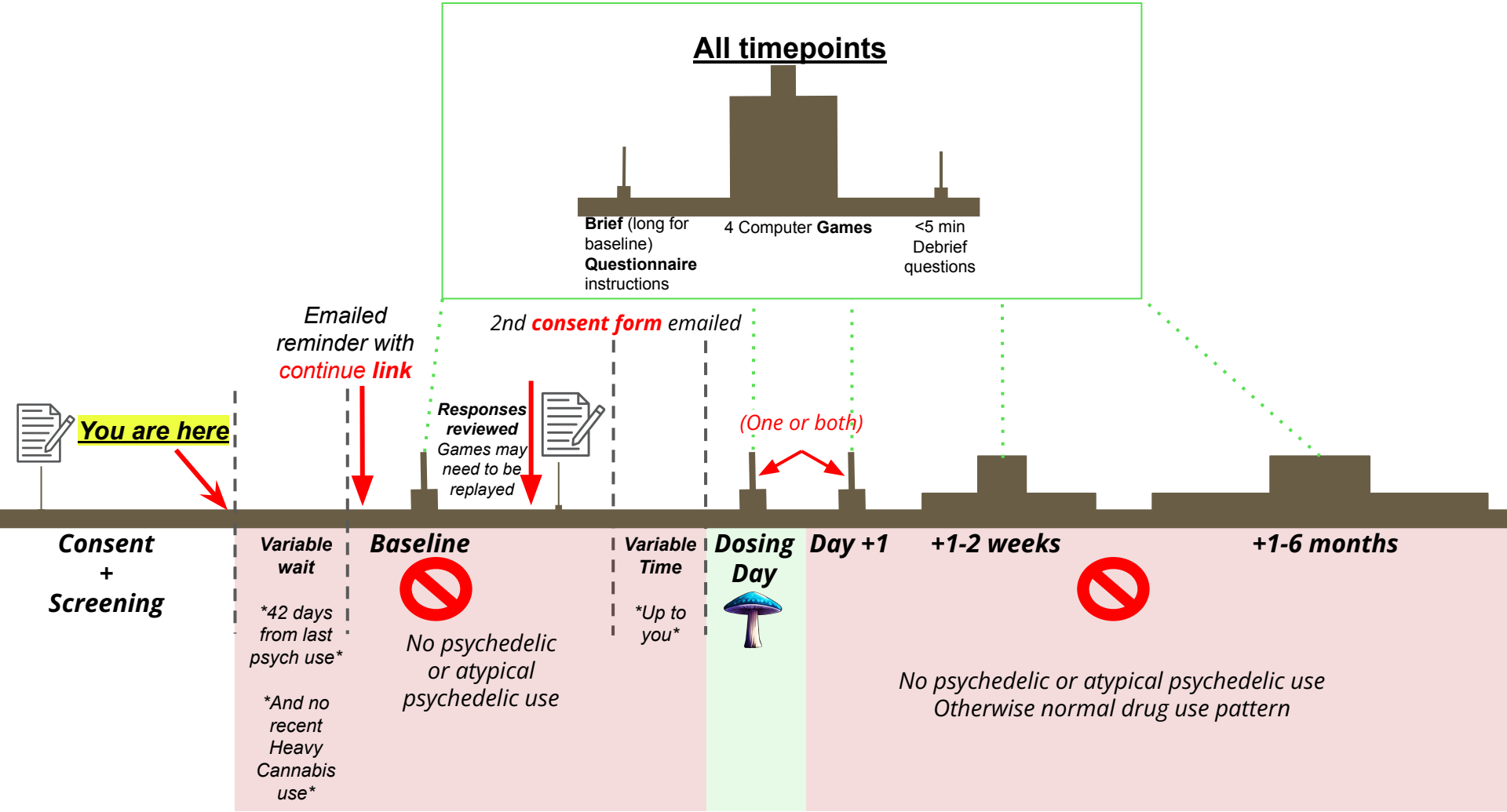
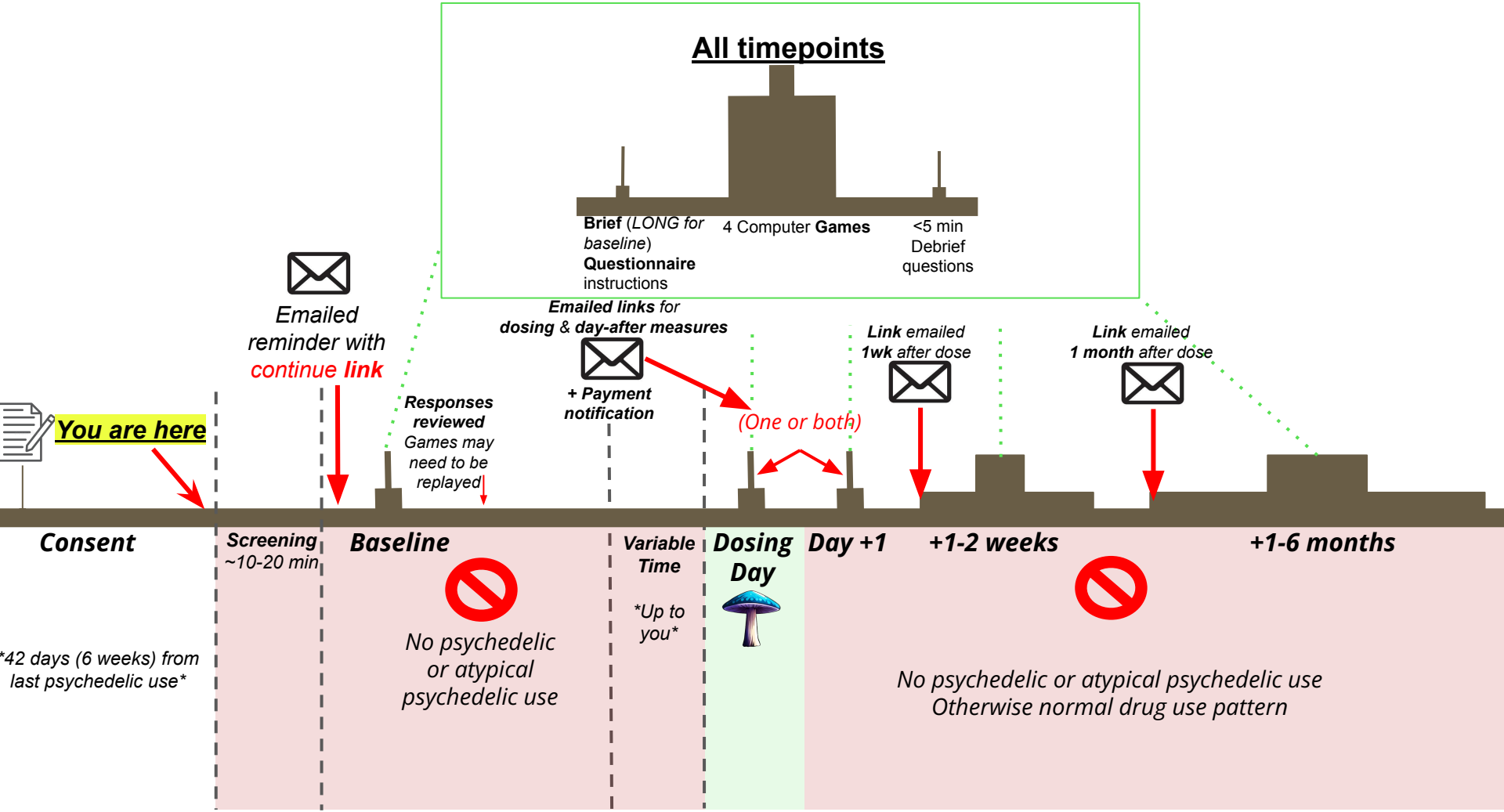
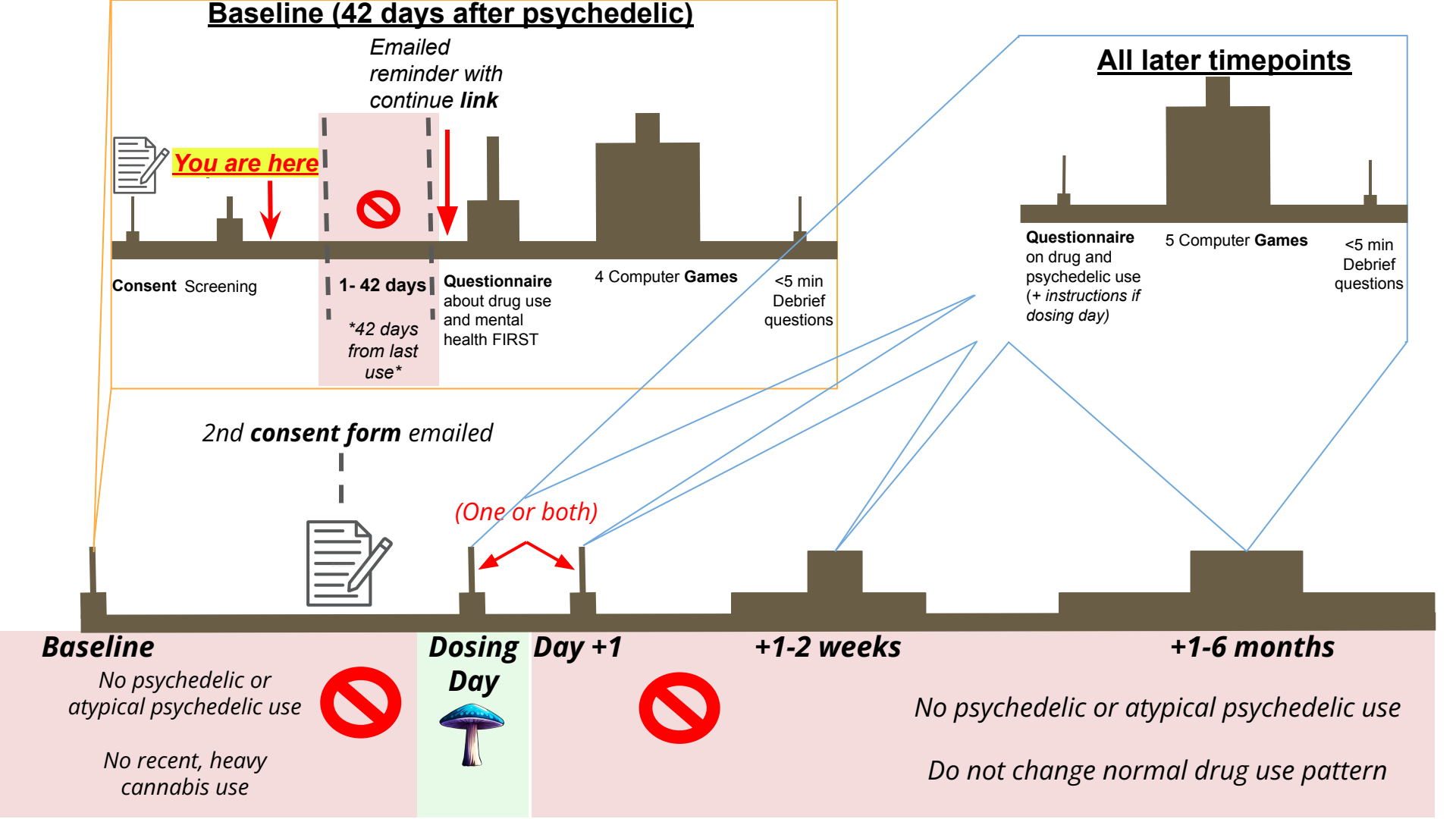
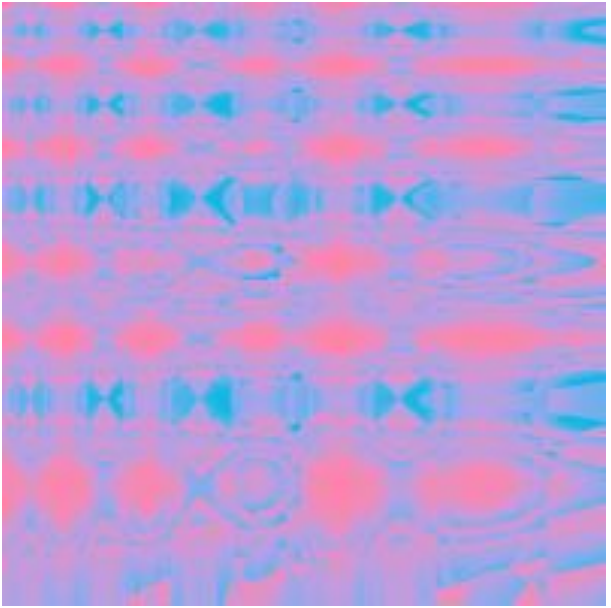
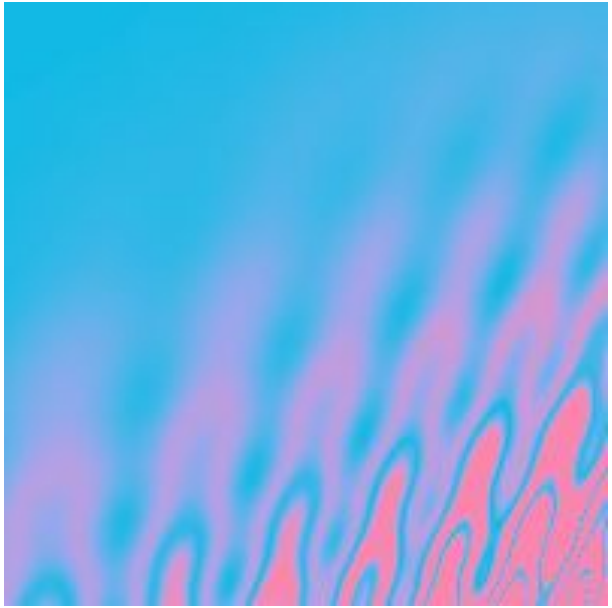

All timepoints
Brief (long for baseline) Questionnaire instructions
4 Computer Games
<5 min Debrief questions
Emailed reminder with continue link
2nd consent form emailed
You are here
Responses reviewed Games may need to be replayed
(One or both)
Consent + Screening
Variable wait
42 days from last psych use
And no recent Heavy Cannabis use
Baseline
No psychedelic or atypical psychedelic use
Variable Time
Up to you
Dosing Day
Day +1
+1-2 weeks
+1-6 months
No psychedelic or atypical psychedelic use
Otherwise normal drug use pattern


All timepoints
Brief (LONG for baseline) Questionnaire instructions
4 Computer Games
<5 min Debrief questions
Emailed reminder with continue link
Emailed links for dosing & day-after measures
+ Payment notification
Responses reviewed Games may need to be replayed
(One or both)
Link emailed 1wk after dose
Link emailed 1 month after dose
You are here
Consent
Screening ~10-20 min
Baseline
Variable Time
Dosing Day
Day +1
+1-2 weeks
+1-6 months
42 days (6 weeks) from last psychedelic use
No psychedelic or atypical psychedelic use
Up to you
No psychedelic or atypical psychedelic use
Otherwise normal drug use pattern

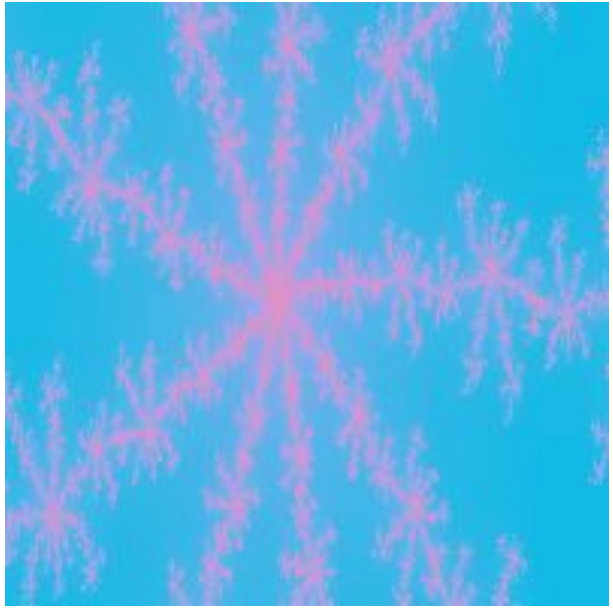

Baseline (42 days after psychedelic)
Emailed reminder with continue link
You are here
Consent Screening
1- 42 days
42 days from last use
Questionnaire about drug use and mental health FIRST
4 Computer Games
<5 min Debrief questions
All later timepoints
Questionnaire on drug and psychedelic use (+ instructions if dosing day)
5 Computer Games
<5 min Debrief questions
2nd consent form emailed
(One or both)
Baseline
No psychedelic or atypical psychedelic use
No recent, heavy cannabis use
Dosing Day
Day +1
+1-2 weeks
+1-6 months
No psychedelic or atypical psychedelic use
Do not change normal drug use pattern



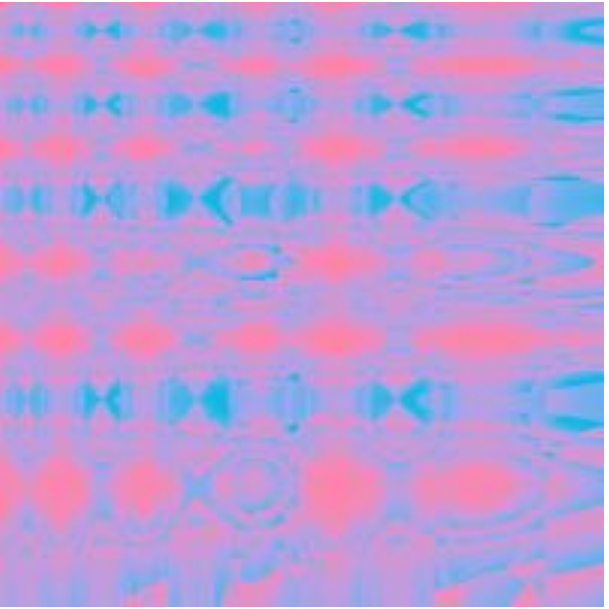
“1”
Left

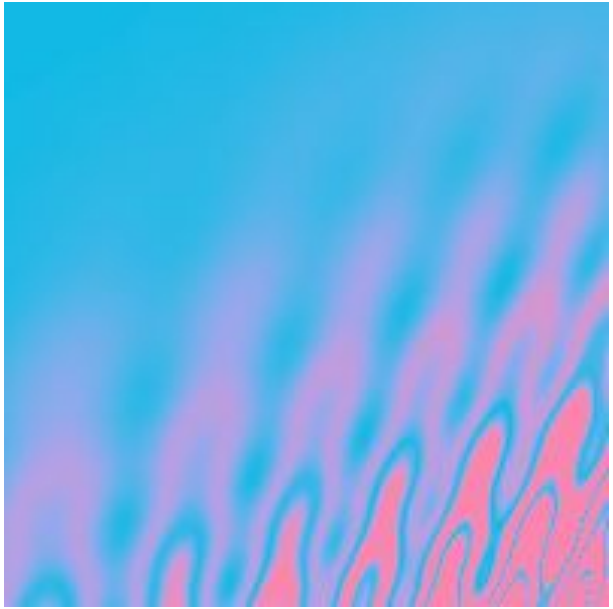


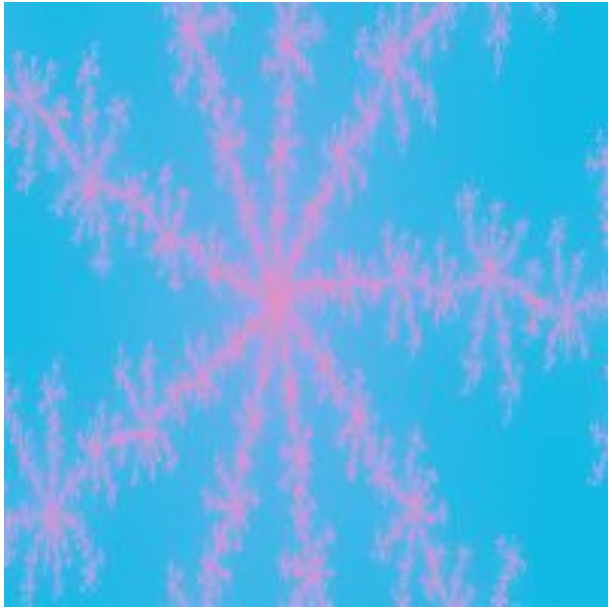
“2”
Middle



“3”
Right







“1”
Left

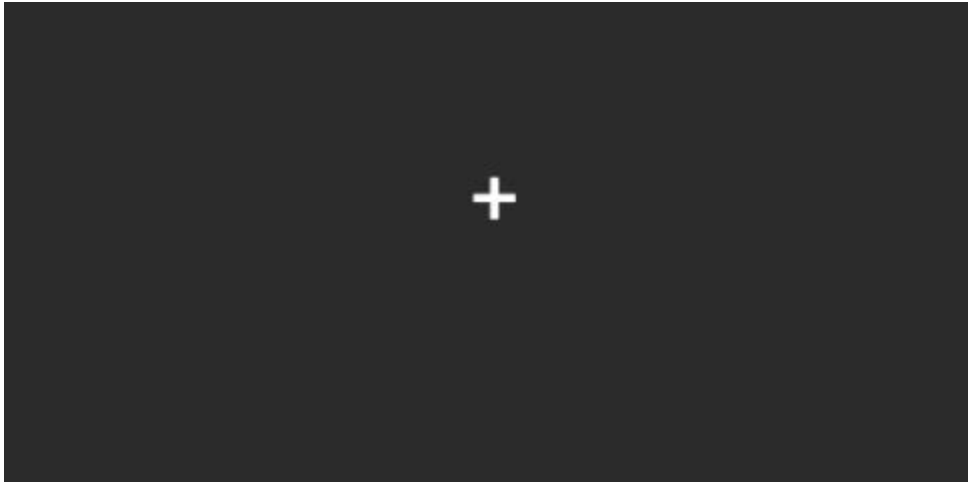
“2”
Middle

“3”
Right

You get 2 points

Total Points: 78

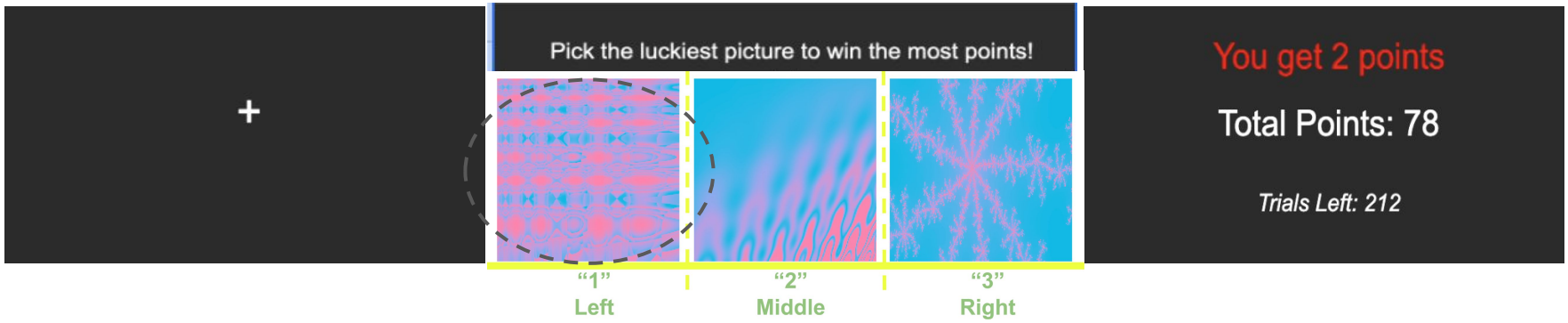
Trials Left: 212



You get 10 points

Total Points: 80

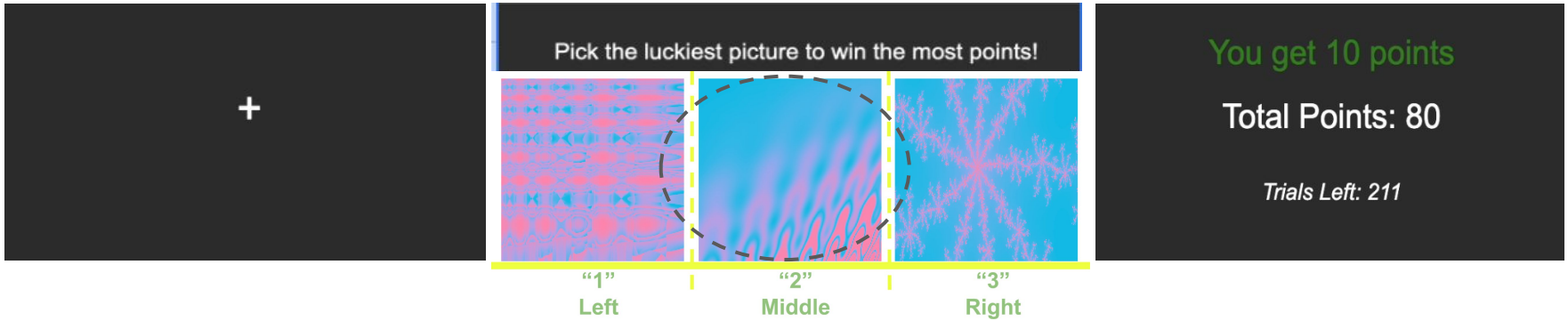
Trials Left: 211



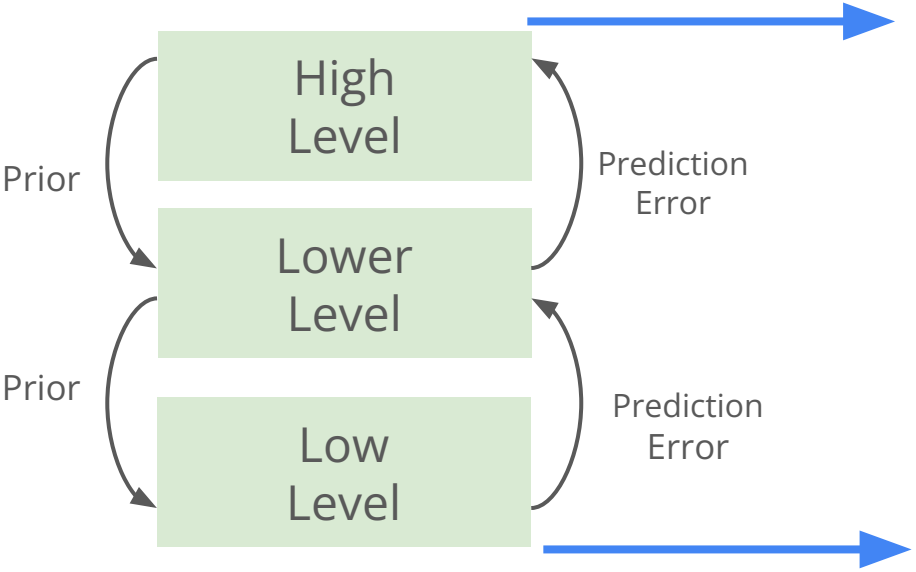
Intertrial interval = 500ms

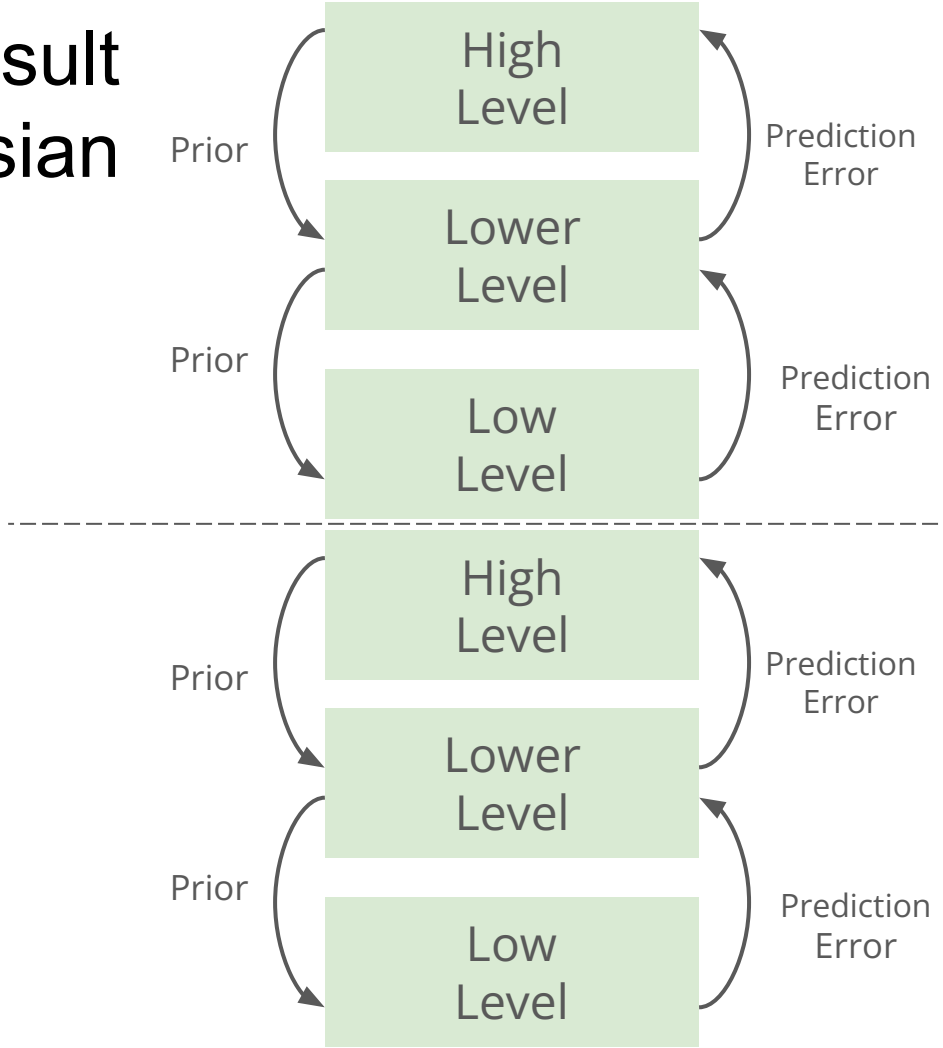
Time to respond = 3000ms (min 200ms)

Feedback display = 1000ms



Psychotic symptoms may result from derangements in bayesian inference

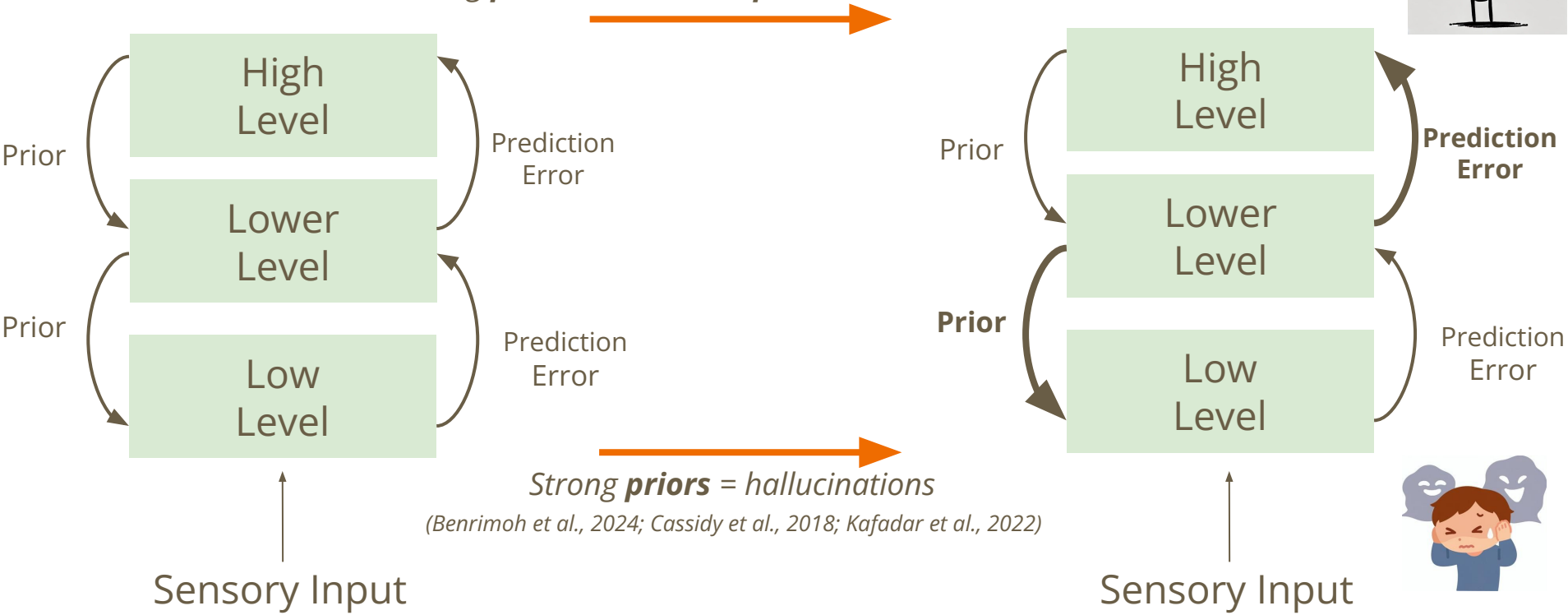




Psychotic symptoms may result from derangements in bayesian inference

(Corlett and Fletcher 2012; Stuke et al. 2019; Corlett et al. 2007)

*Strong **prediction errors (pwPEs)** = aberrant salience*

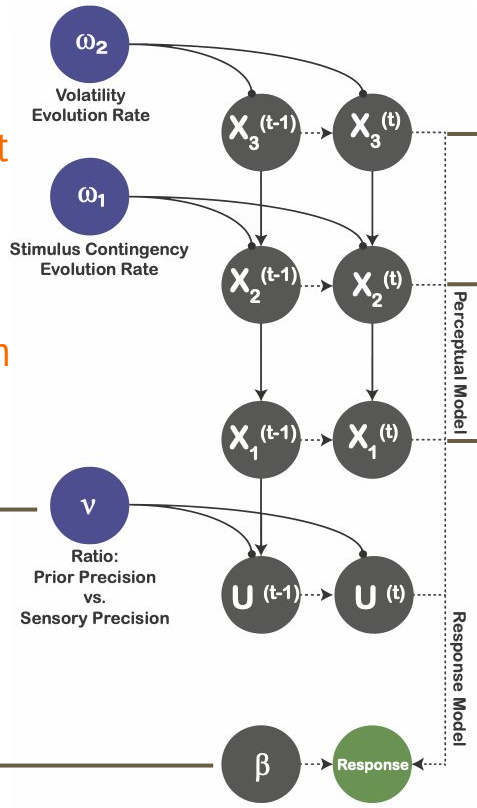


*Strong **priors** = hallucinations*

(Benrimoh et al., 2024; Cassidy et al., 2018; Kafadar et al., 2022)

Hierarchical Gaussian Filter (HGF) modelling of CH Task estimates bayesian inference at multiple belief levels

The **HGF** consists of a generative perceptual and decision model explaining how **high level beliefs** about perceptual patterns (eg. how liable associations are to change) influence **low level perceptual inference** (whether a tone is present) and result in participant **choices** (eg. in CH task)



*Belief that visual-auditory stimulus association **can change** (volatility)*

*Overall belief that visual stimulus predicts auditory tone (the **prior**)*

*Trial-wise **belief in tone presence** — expressed as probability*

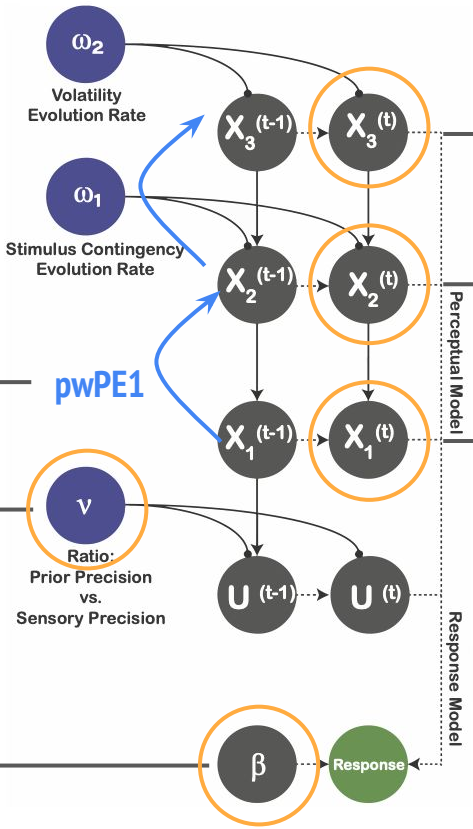
***Prior weighting** compared to sensory evidence (tone **volume**) weighting*

***Decision noise** (errors in detection indication)*

The HGF is **inverted** from participant data to estimate **each participant's beliefs and prior weighting**. (Mathys et al. 2014)

Measures - Computational (HGF)

Parameters estimated for each individual based on trial-by-trial responses and stimulus intensity using MATlab HGF toolbox



pwPE2

pwPE1

*Belief that visual-auditory stimulus association **can change** (volatility)*

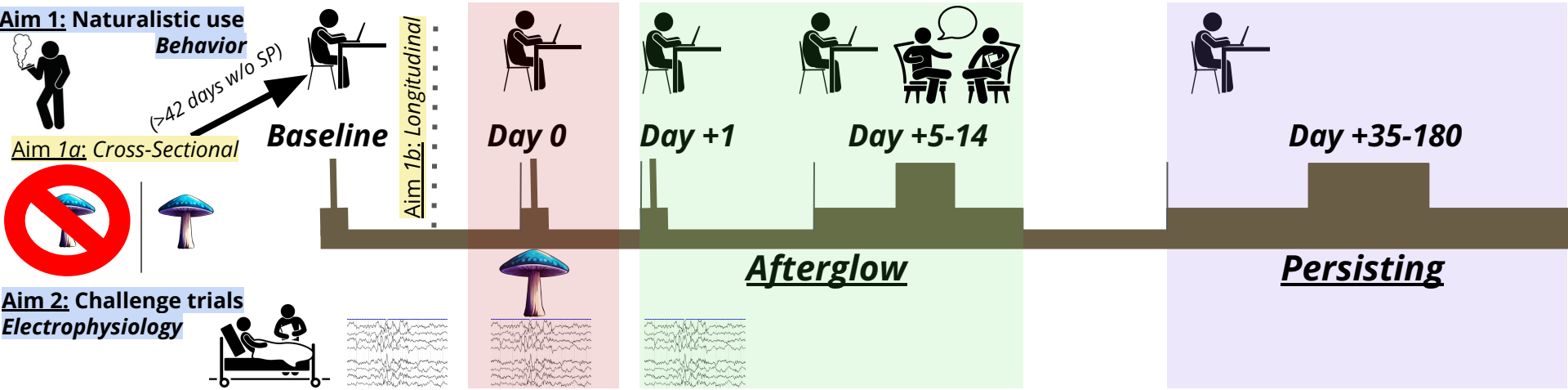
*Overall belief that visual stimulus predicts auditory tone (the **prior**)*

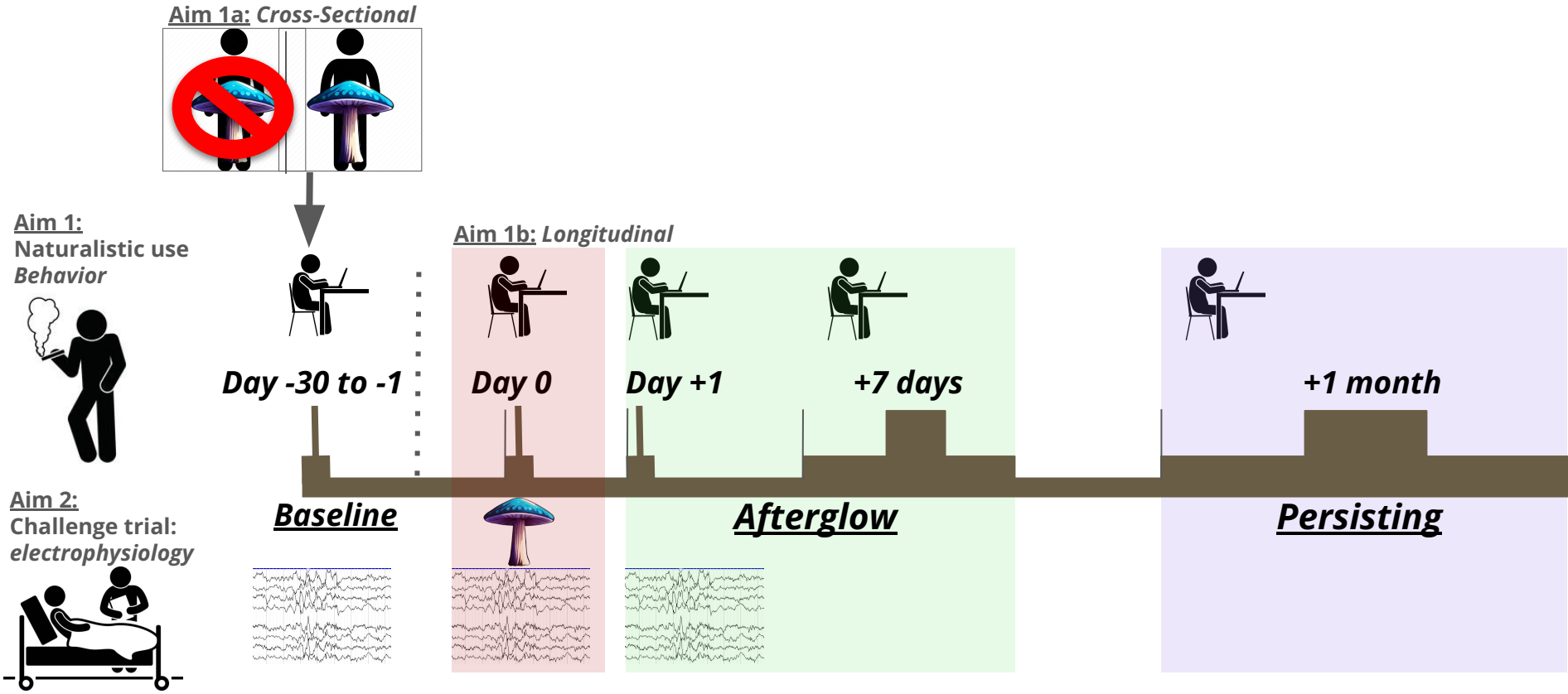
*Trial-wise **belief in tone presence** — expressed as probability*

*Trial wise precision-weighted **prediction error** (weakens pavlovian conditioning as function of contrary sensory evidence!)*

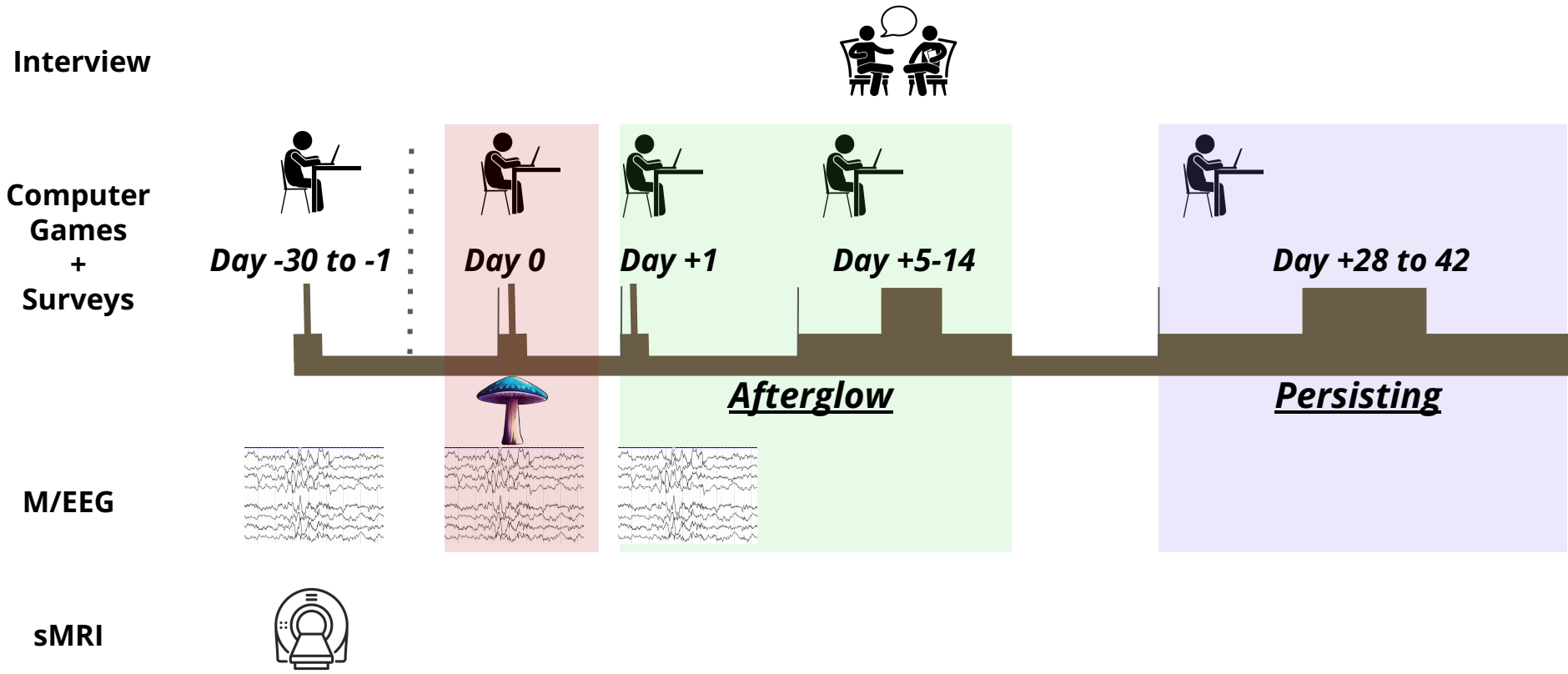
***Prior weighting** compared to sensory evidence (tone **volume**) weighting*

***Decision noise** (errors in detection indication)*

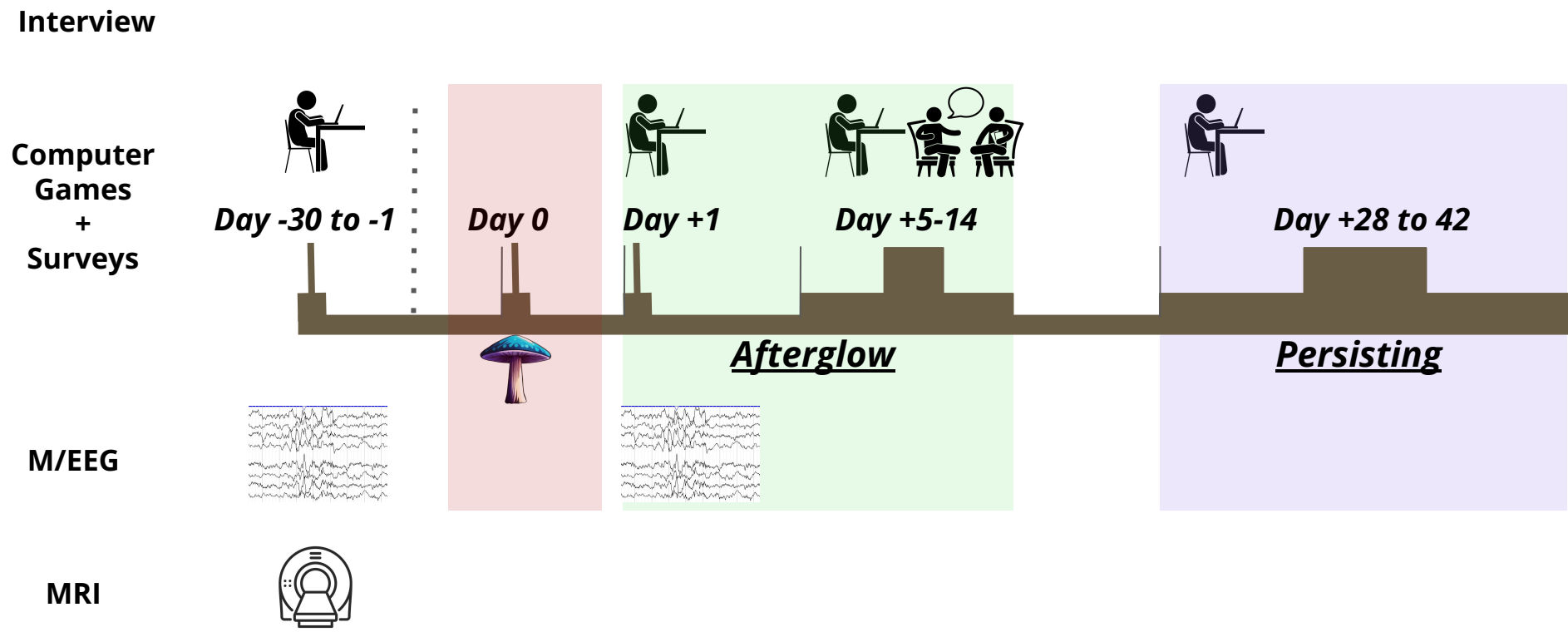

Aim 1: Naturalistic use
Behavior
(>42 days w/o SP)
Aim 1a: Cross-Sectional
Baseline
Aim 1b: Longitudinal
Day 0
Day +1
Day +5-14
Day +35-180
Afterglow
Persisting
Aim 2: Challenge trials
Electrophysiology


Aim 1a: Cross-Sectional
Aim 1:
Naturalistic use
Behavior
Aim 1b: Longitudinal
Day -30 to -1
Day 0
Day +1
+7 days
+1 month
Baseline
Afterglow
Persisting
Aim 2:
Challenge trial:
electrophysiology

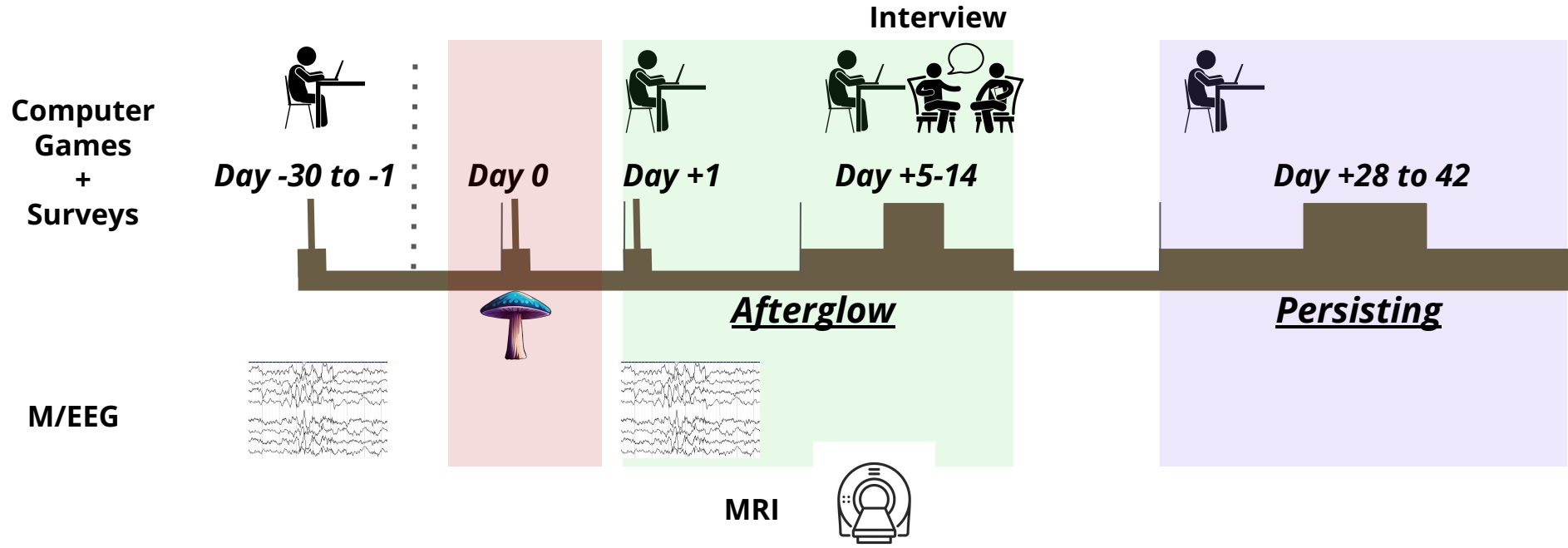
Full Study Version



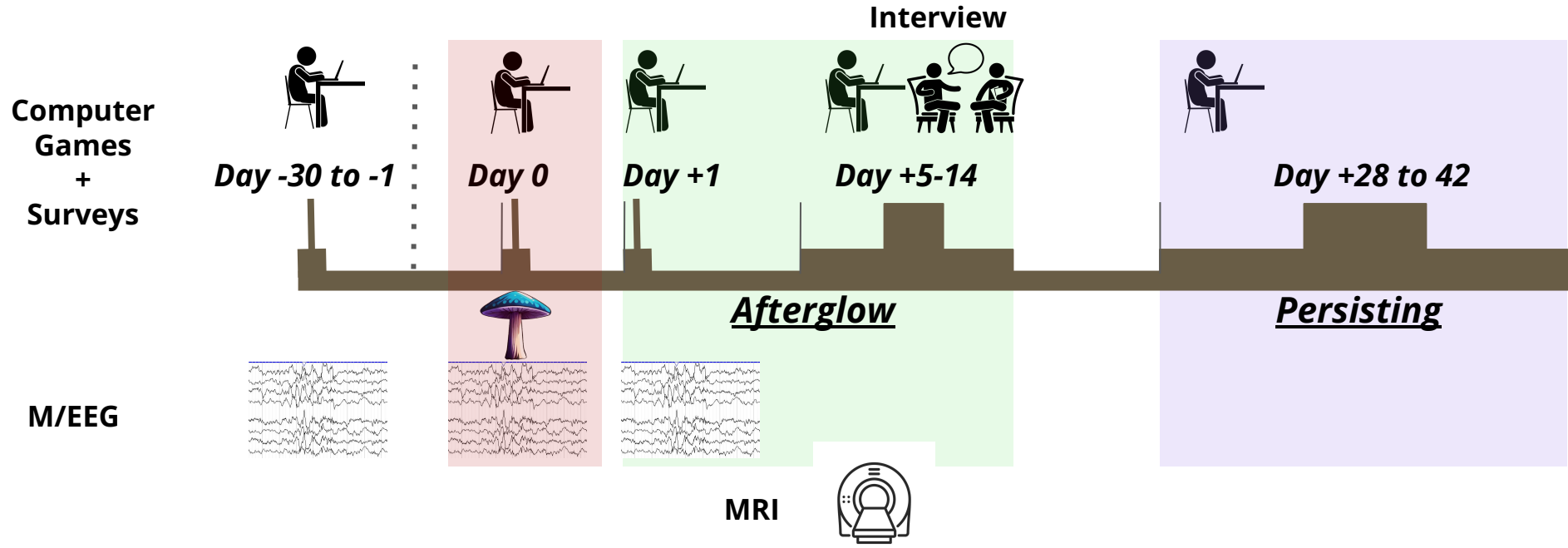
No Day 0 Assessment Version



No Day 0 Assessment Version



Yes Day 0 Assessment Version



View

Sala de reuniones personales de Max Gr...

You to Everyone 3:39 PM

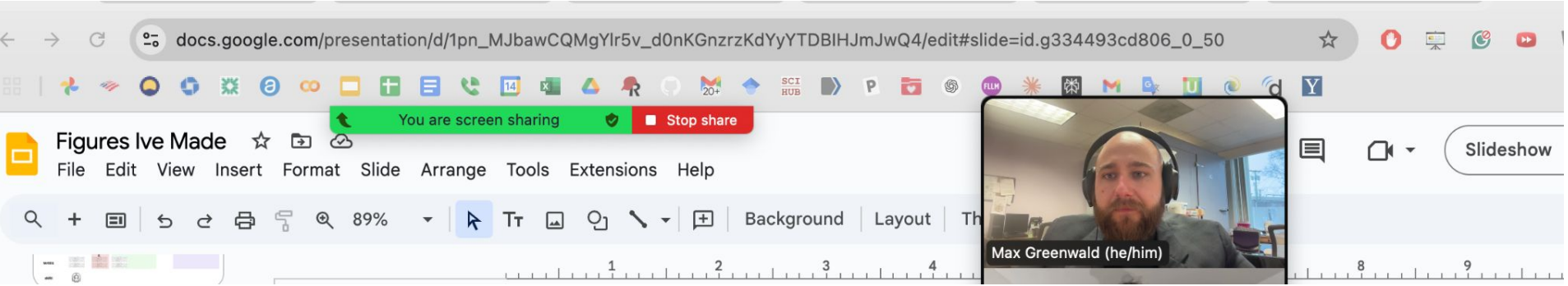
https://redcap.research.yale.edu/surveys/index.php?s=6KNeAXgY6NtkbWFB

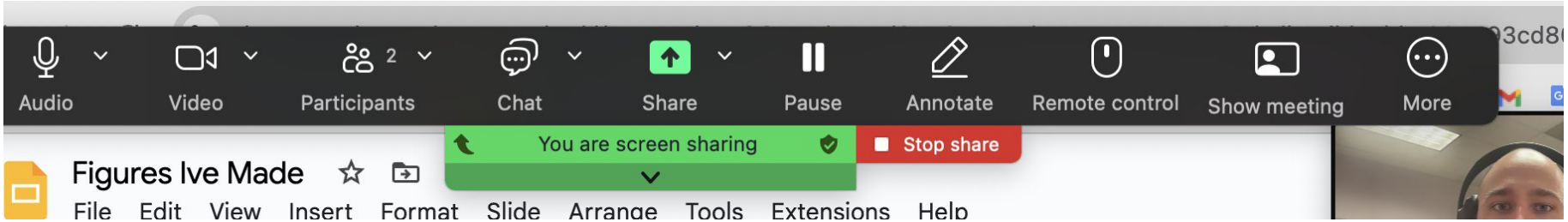
Who can see your messages?

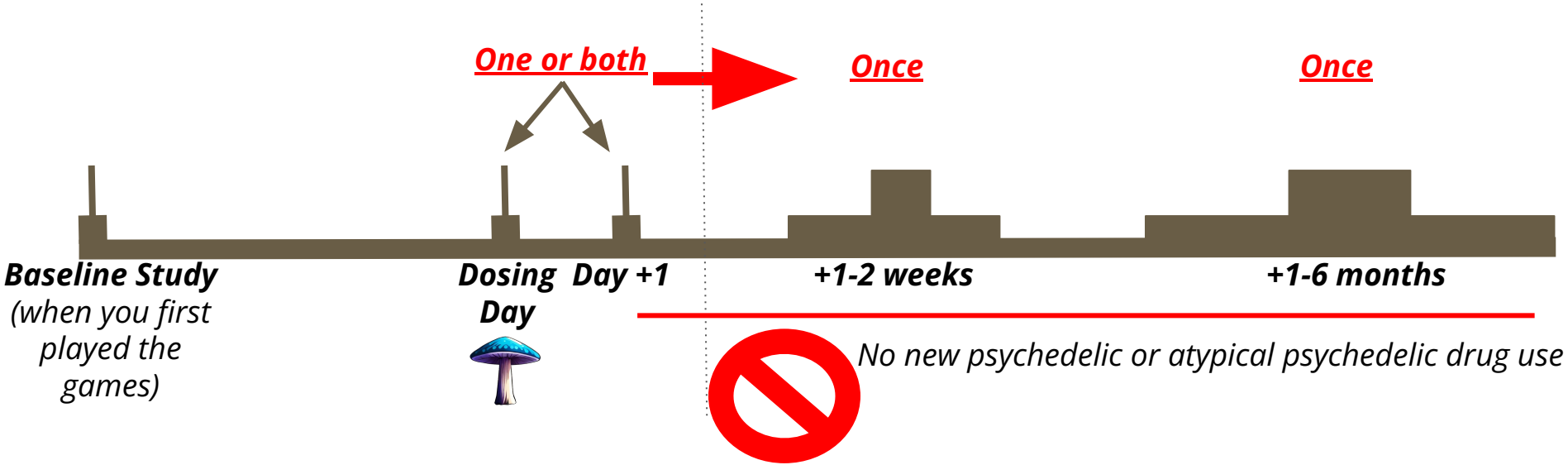
To: Everyone

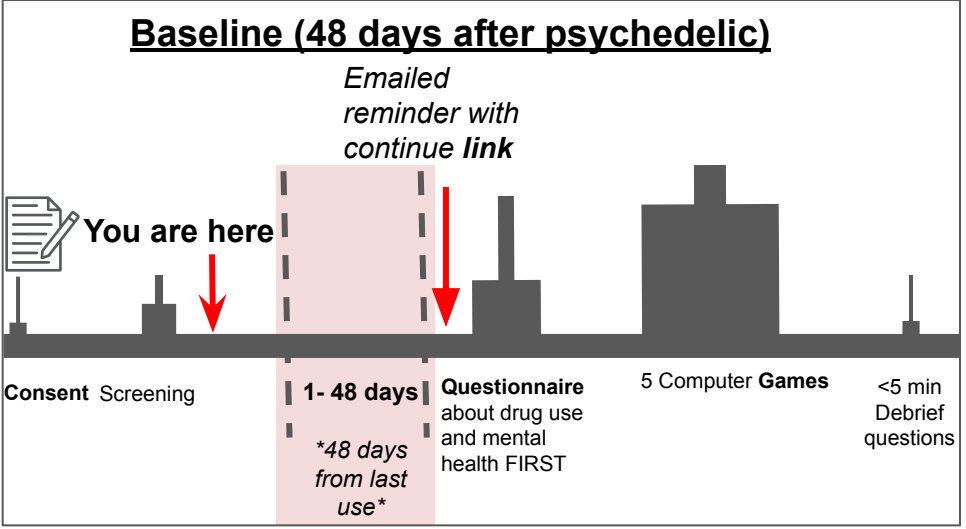
Type message here...

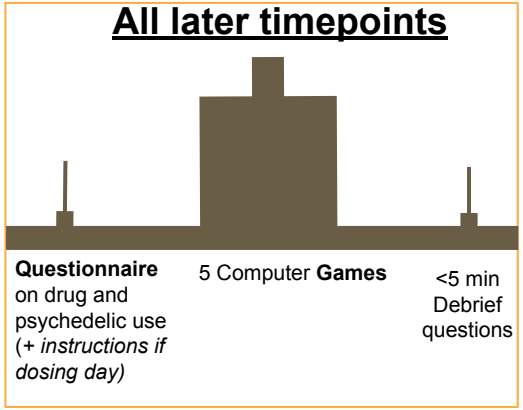
Audio Video Participants 2 Chat React Share (⇧⌘S) st tools AI Companion Record More End

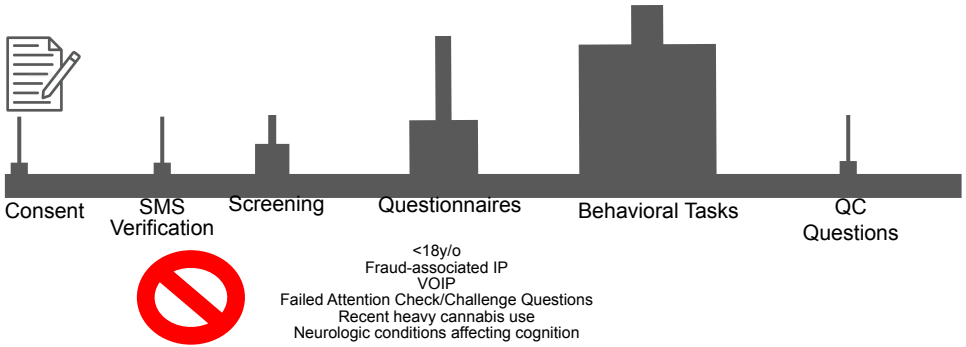

docs.google.com/presentation/d/1pn_MJbawCQMgYlr5v_d0nKGnzrzKdYyYTDBIHJmJwQ4/edit#slide=id.g334493cd806_0_50
You are screen sharing
Stop share
Figures Ive Made
File Edit View Insert Format Slide Arrange Tools Extensions Help
89%
Background
Layout
Slideshow
Max Greenwald (he/him)

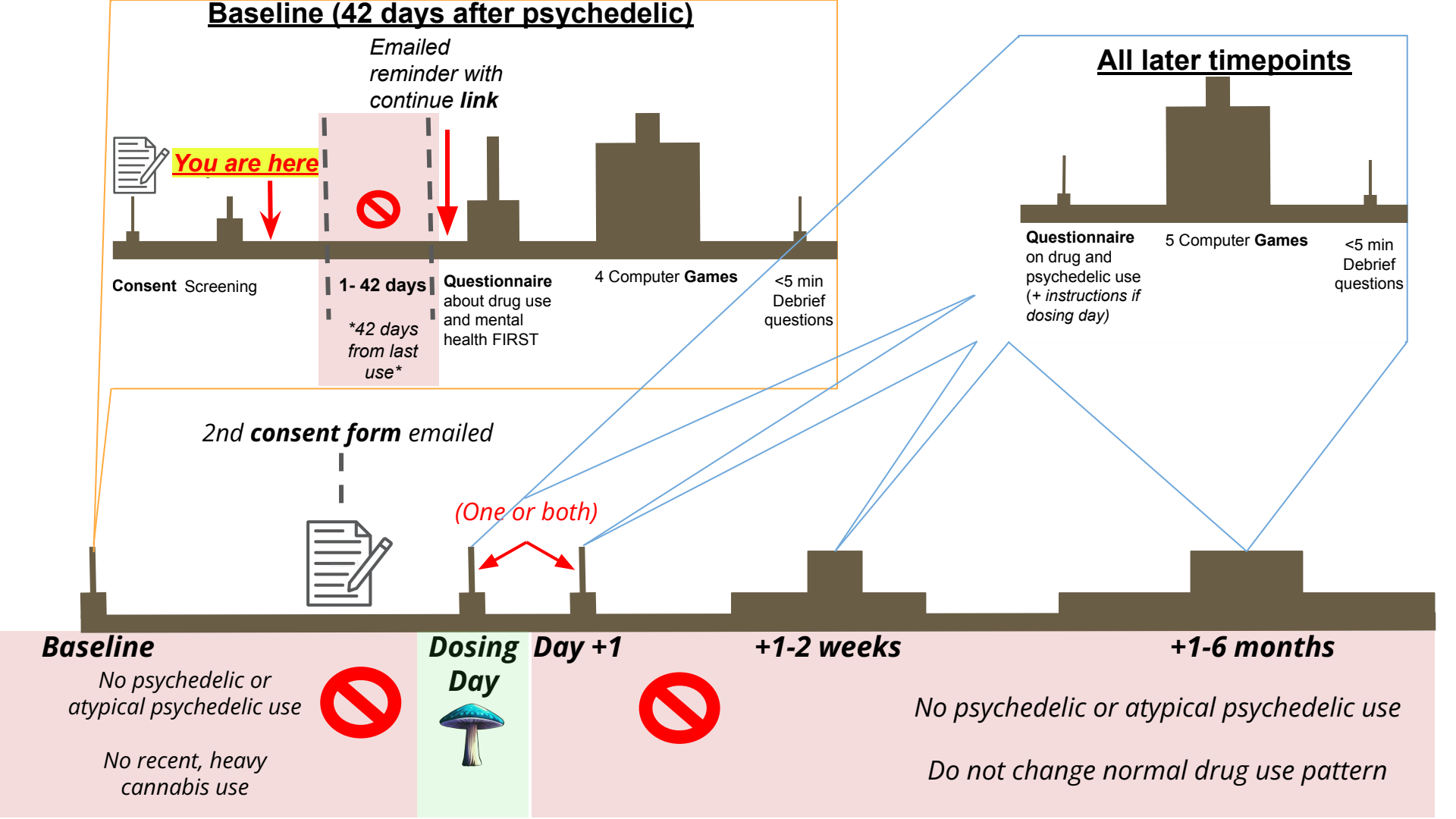

Audio
Video
Participants
2
Chat
Share
Pause
Annotate
Remote control
Show meeting
More
You are screen sharing
Stop share
Figures Ive Made
File Edit View Insert Format Slide Arrange Tools Extensions Help

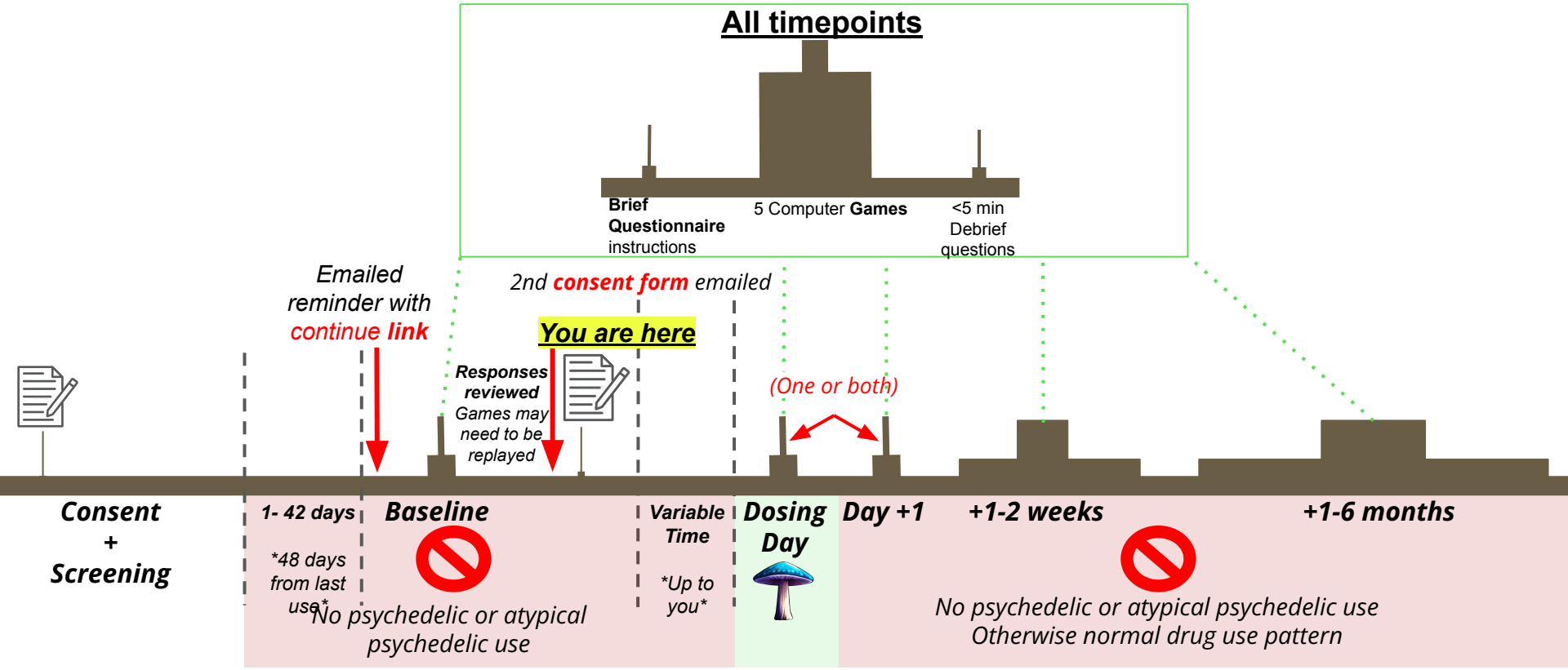

One or both
Once
Once
Baseline Study
(when you first played the games)
Dosing Day
Day +1
+1-2 weeks
+1-6 months
No new psychedelic or atypical psychedelic drug use

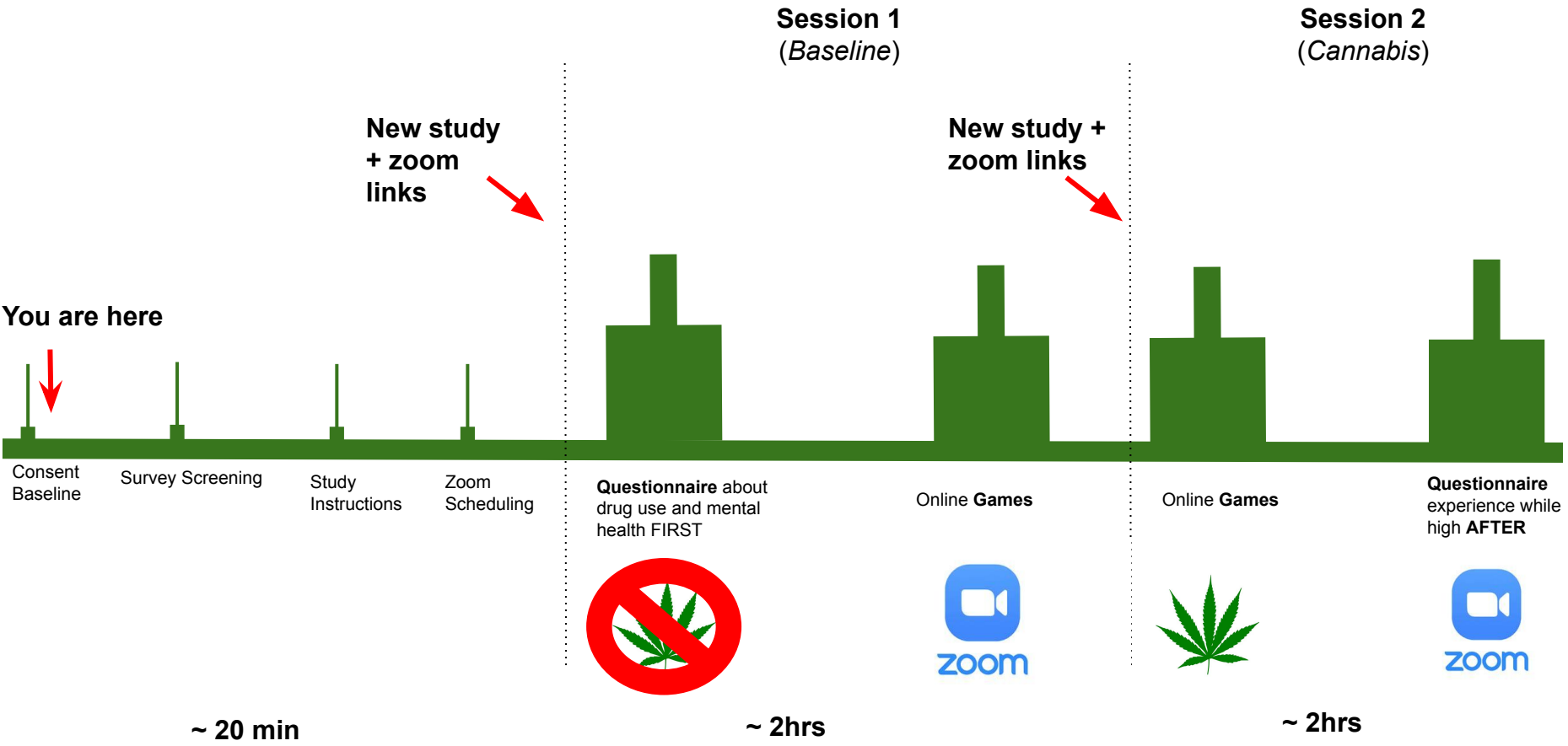

Baseline (48 days after psychedelic)
Emailed reminder with continue link
You are here
Consent Screening
1- 48 days
48 days from last use
Questionnaire about drug use and mental health FIRST
5 Computer Games
<5 min Debrief questions

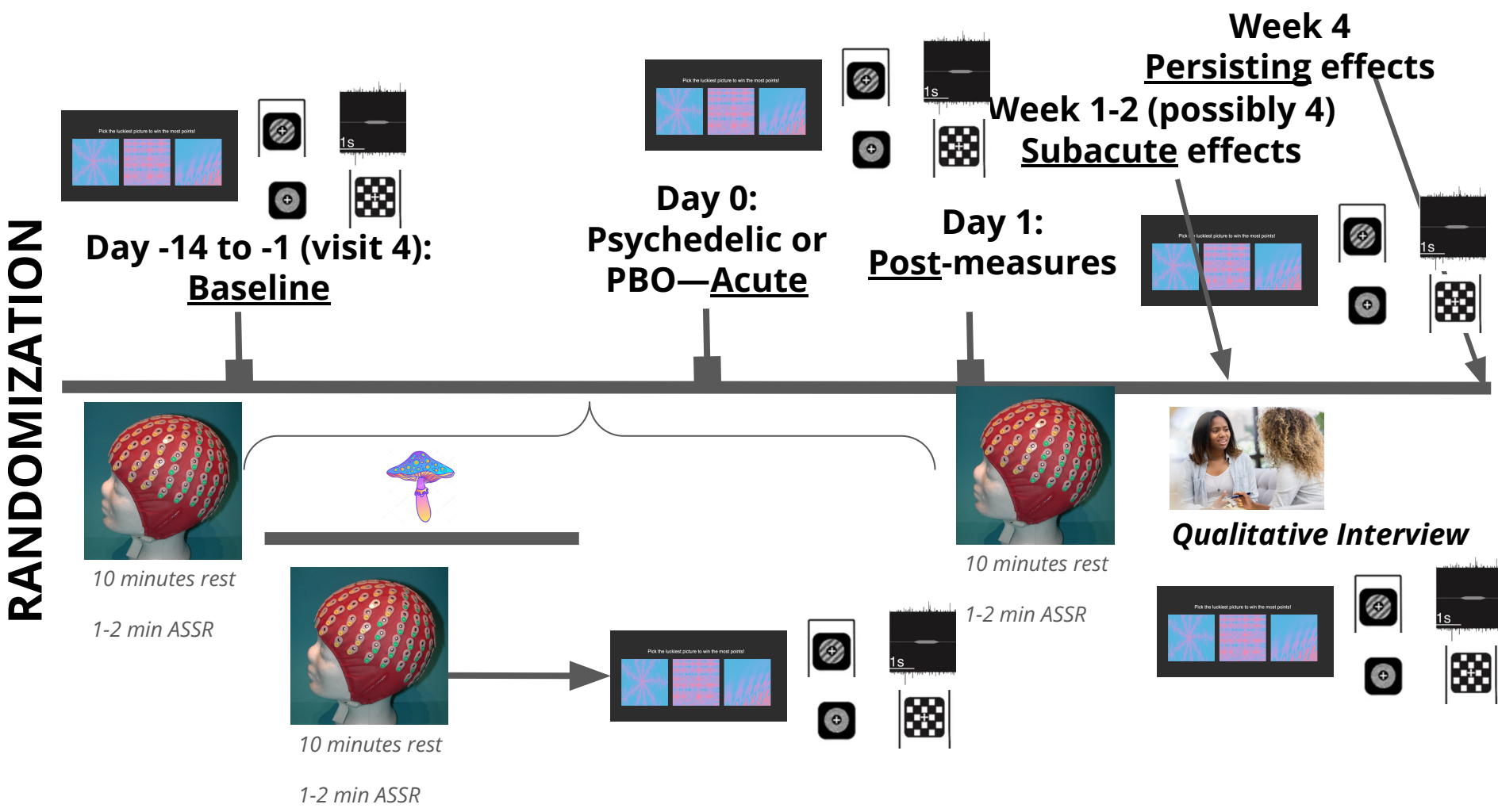

All later timepoints
Questionnaire on drug and psychedelic use (+ instructions if dosing day)
5 Computer Games
<5 min Debrief questions

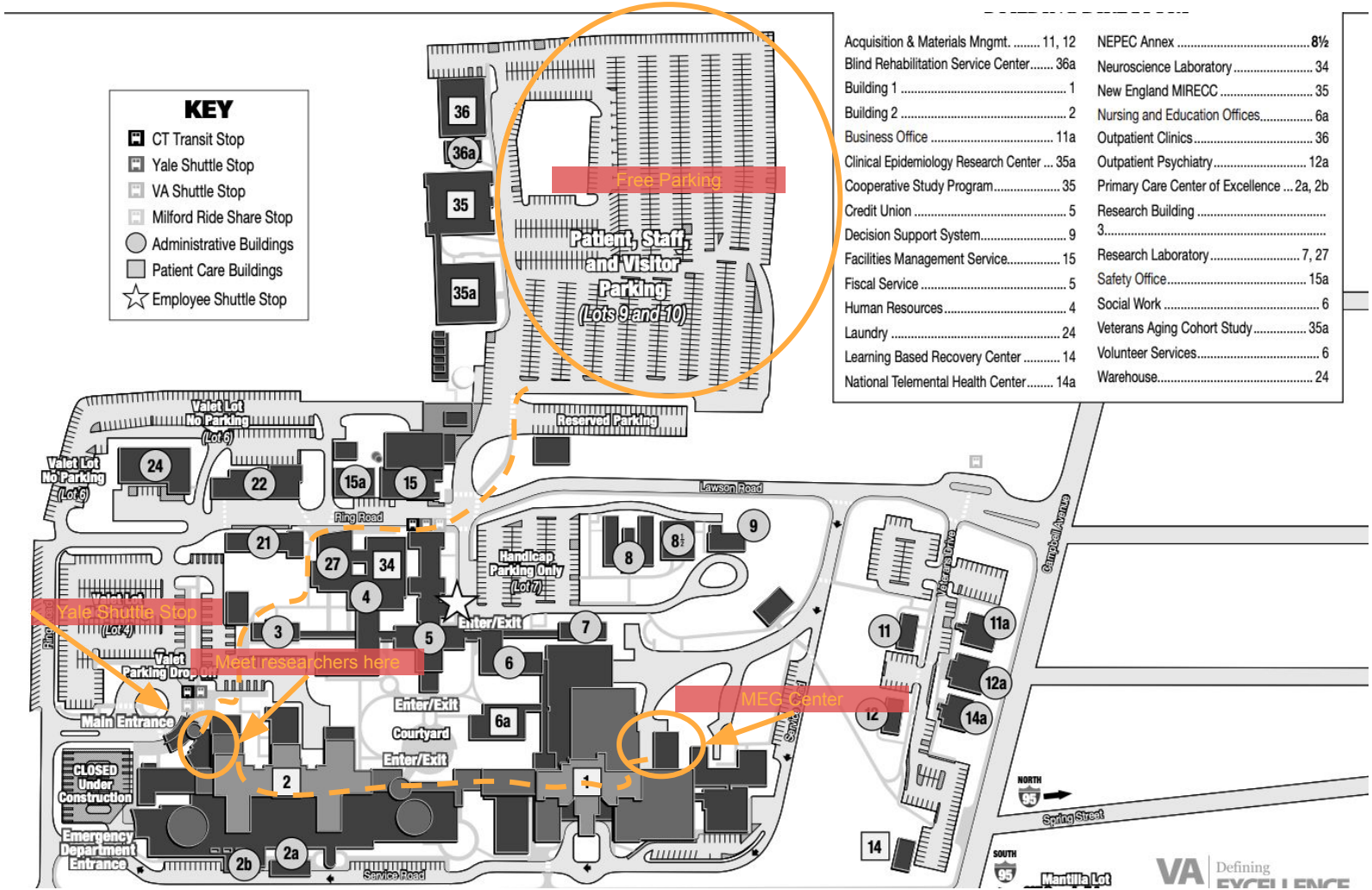

Consent
SMS Verification
Screening
Questionnaires
Behavioral Tasks
QC Questions
<18y/o
Fraud-associated IP
VOIP
Failed Attention Check/Challenge Questions
Recent heavy cannabis use
Neurologic conditions affecting cognition


Baseline (42 days after psychedelic)
Emailed reminder with continue link
You are here
Consent Screening
1- 42 days
42 days from last use
Questionnaire about drug use and mental health FIRST
4 Computer Games
<5 min Debrief questions
All later timepoints
Questionnaire on drug and psychedelic use (+ instructions if dosing day)
5 Computer Games
<5 min Debrief questions
2nd consent form emailed
(One or both)
Baseline
No psychedelic or atypical psychedelic use
No recent, heavy cannabis use
Dosing Day
Day +1
+1-2 weeks
+1-6 months
No psychedelic or atypical psychedelic use
Do not change normal drug use pattern


All timepoints
Brief Questionnaire instructions
5 Computer Games
<5 min Debrief questions
Emailed reminder with continue link
2nd consent form emailed
You are here
Responses reviewed
Games may need to be replayed
(One or both)
Consent + Screening
1- 42 days
48 days from last use
Baseline
No psychedelic or atypical psychedelic use
Variable Time
Up to you
Dosing Day
Day +1
+1-2 weeks
+1-6 months
No psychedelic or atypical psychedelic use
Otherwise normal drug use pattern


Session 1
(Baseline)
Session 2
(Cannabis)
New study + zoom links
New study + zoom links
You are here
Consent Baseline
Survey Screening
Study Instructions
Zoom Scheduling
Questionnaire about drug use and mental health FIRST
Online Games
Online Games
Questionnaire experience while high AFTER
zoom
zoom
~ 20 min
~ 2hrs
~ 2hrs


RANDOMIZATION
Day -14 to -1 (visit 4):
Baseline
Day 0:
Psychedelic or
PBO—Acute
Day 1:
Post-measures
Week 1-2 (possibly 4)
Subacute effects
Week 4
Persisting effects
Qualitative Interview
10 minutes rest
1-2 min ASSR
10 minutes rest
1-2 min ASSR
10 minutes rest
1-2 min ASSR
Pick the luckiest picture to win the most points!
1s


KEY
CT Transit Stop
Yale Shuttle Stop
VA Shuttle Stop
Milford Ride Share Stop
Administrative Buildings
Patient Care Buildings
Employee Shuttle Stop
Free Parking
Patient, Staff, and Visitor Parking (Lots 9 and 10)
Reserved Parking
Valet Lot No Parking (Lot 6)
Valet Lot No Parking (Lot 5)
Ring Road
Handicap Parking Only (Lot 7)
Enter/Exit
Yale Shuttle Stop
Meet researchers here
MEG Center
Valet Parking Drop Off
Main Entrance
Courtyard
CLOSED Under Construction
Emergency Department Entrance
Service Road
Lawson Road
Campbell Avenue
Veterans Drive
Spring Street
NORTH 95
SOUTH 95
Mantilla Lot
36
36a
35
35a
24
22
15a
15
21
27
34
4
3
5
6
6a
7
8
8½
9
11
11a
12
12a
14
14a
2
2a
2b
1
Acquisition & Materials Mngmt. 11, 12
Blind Rehabilitation Service Center..... 36a
Building 1 1
Building 2 2
Business Office 11a
Clinical Epidemiology Research Center ... 35a
Cooperative Study Program..... 35
Credit Union 5
Decision Support System..... 9
Facilities Management Service..... 15
Fiscal Service 5
Human Resources..... 4
Laundry 24
Learning Based Recovery Center 14
National Telemental Health Center..... 14a
NEPEC Annex 8½
Neuroscience Laboratory..... 34
New England MIRECC 35
Nursing and Education Offices..... 6a
Outpatient Clinics..... 36
Outpatient Psychiatry..... 12a
Primary Care Center of Excellence ... 2a, 2b
Research Building ..... 3.....
Research Laboratory..... 7, 27
Safety Office..... 15a
Social Work 6
Veterans Aging Cohort Study..... 35a
Volunteer Services..... 6
Warehouse..... 24
VA Defining EXCELLENCE

Yale shuttle stop
Main Entrance
Meet researcher just past doors
ACCESS
Building Two

Shuttle Routes

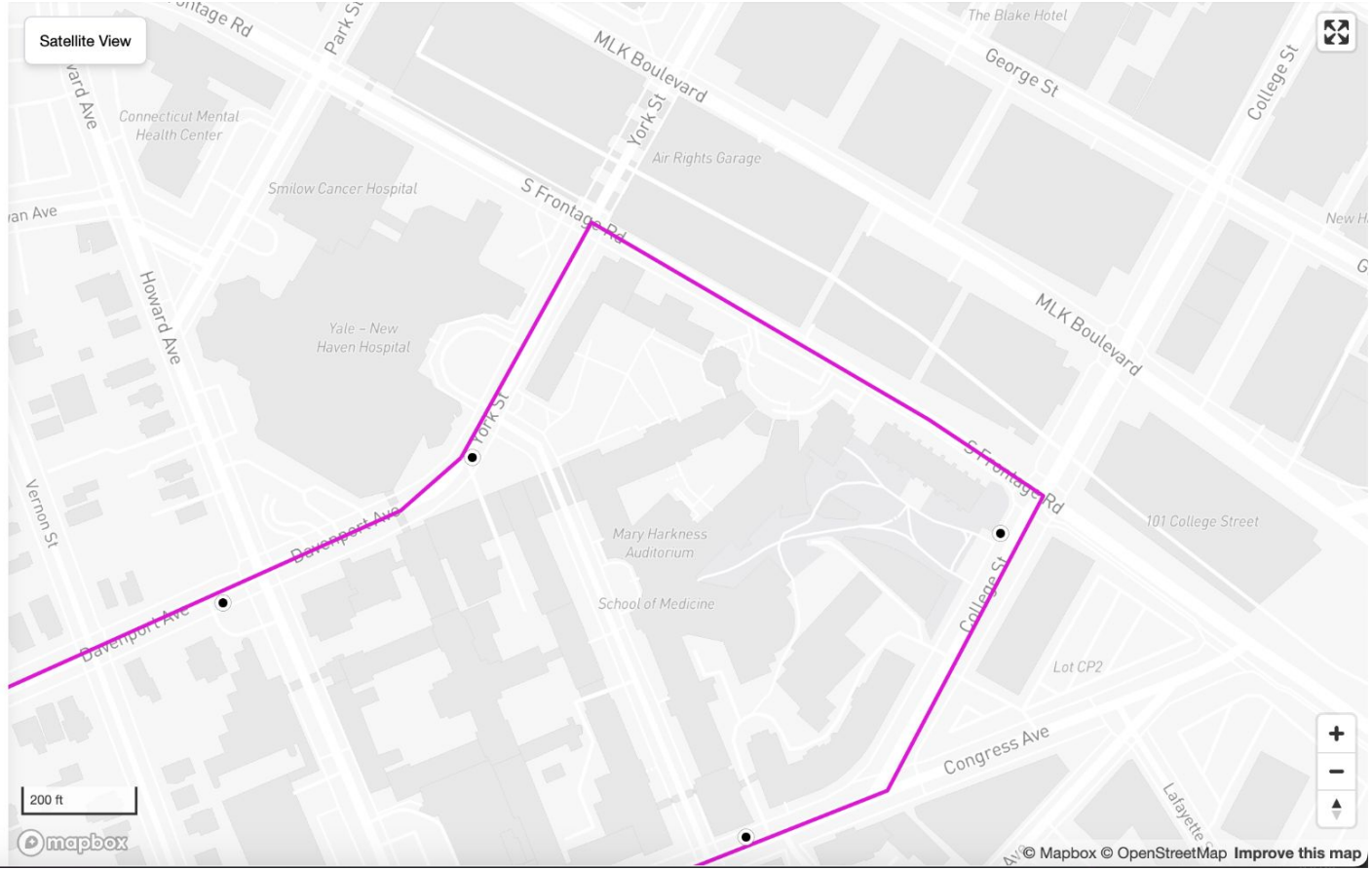
All | None

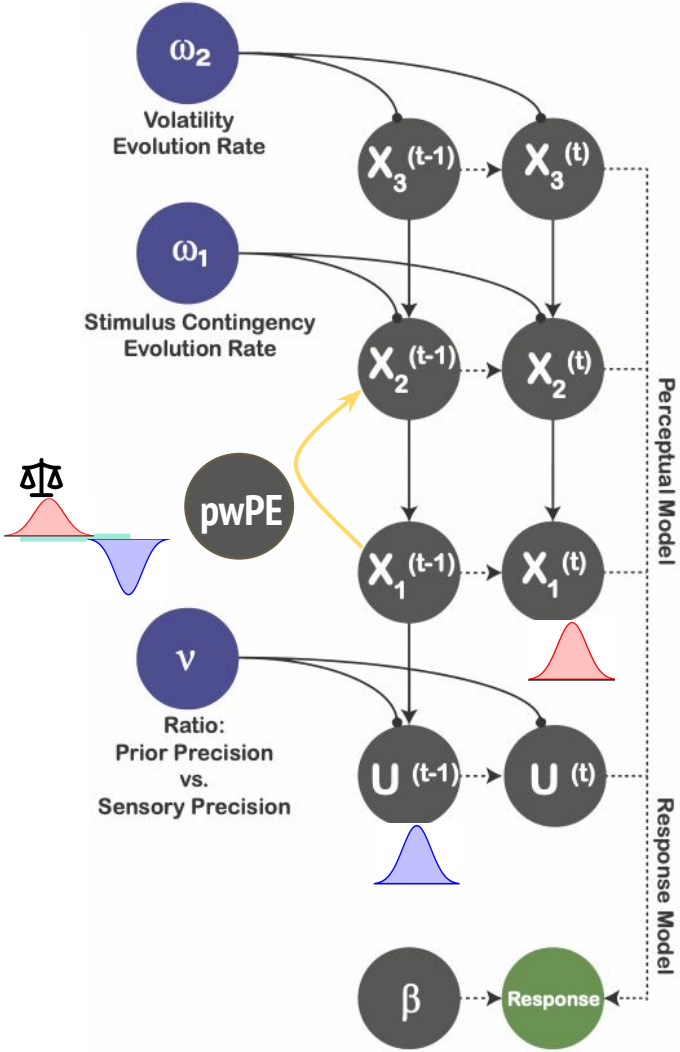
ONLINE

- **Blue - Weekday Daytime**
  7am - 6pm, M - F
- **Orange - Weekday Daytime**
  7am - 6pm, M - F
- **Red - Weekday Daytime**
  7am - 6pm, M - F
- **Pink - VA Hospital / Med School**
  6am - 6:30pm, M - F
- **Green - West Campus**
  5:20am - 6:40pm, Daily
- **Purple - West Campus**
  5:30am - 11:45pm, Daily
- **Brown - Science Park Garage**
  6am - 7pm, M - F
- **Yellow - Central Campus Train**
  6am - 7pm, M - F

OFFLINE

- **Blue - Weekend Daytime**
  7am - 6:20pm, Sat - Sun
- **Weekend Grocery**
  7am - 5pm, Sat - Sun
- **Cedar Express**
  6pm - 1am, Daily




ω2
Volatility
Evolution Rate
ω1
Stimulus Contingency
Evolution Rate
pwPE
ν
Ratio:
Prior Precision
vs.
Sensory Precision
X3 (t-1)
X3 (t)
X2 (t-1)
X2 (t)
X1 (t-1)
X1 (t)
U (t-1)
U (t)
β
Response
Perceptual Model
Response Model